EIH Camera API Function Manual

Original Instructions

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TECHMAN ROBOT INC.
5F., No. 58-2, Huaya 2nd Rd., Guishan Dist., Taoyuan City, 333411 , Taiwan

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Revision History

Revision	Date	Revised Content
1.0	2024-11-07	Original release

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1. Overview

Introducing the EIH Camera API, a gRPC-based interface designed for the embedded Eye-in-Hand (EIH) camera on TM robots. With this powerful API, developers can read and write camera parameters such as Focus Working Range, Image Size, and Distortion Coefficients, offering precise control over the camera and enabling developers to integrate these images with their algorithms to build advanced and customized vision systems. The EIH Camera API opens up new possibilities for creating customized, high-performance vision solutions that enhance the capabilities of your automation solutions.

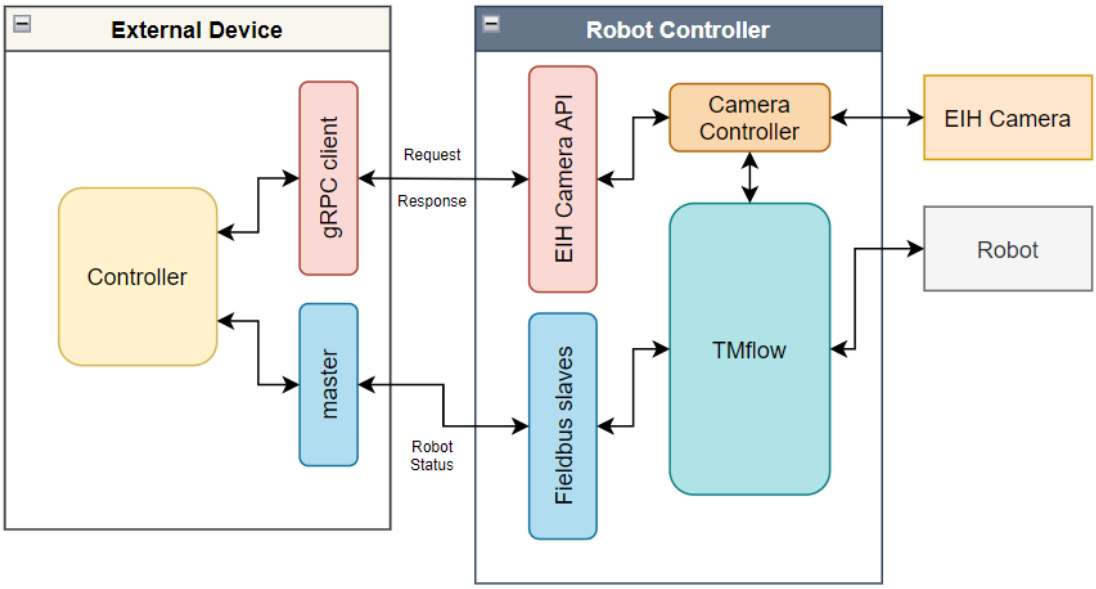

Figure 1: EIH Camera API Architecture

With EIH Camera API, users can request the following data:

- ✓ Camera Matrix
- ✓ Distortion Coefficients
- ✓ Hand Eye Array
- ✓ White Balance
- ✓ Pixel Format
- ✓ Image Size
- ✓ Image Type
- ✓ Focus Value
- ✓ Shuttle Time
- ✓ Gain

Before using the EIH Camera API, developers should have the following prerequisites:

- ✓ Knowledge of gRPC (e.g. building a gRPC client)
- ✓ Knowledge of using TMflow and TMvision

2. Enabling EIH Camera API Server

To use the EIH Camera API Server, users must first enable the server by navigating to **Configuration** > **Vision Settings** > **Service** > **EIH Camera** in TMflow.

If needed, users can set up an IP accept list of IPs using the IP filter function for enhanced security. Finally, click the Enable button to start the EIH Camera API Server (**port: 15567**).

IMPORTANT

IMPORTANT:
If TMvision is running a vision job while the EIH Camera API requests the EIH Camera simultaneously, TMvision will always take priority. The EIH Camera API will remain waiting until the camera is released.

3. EIH Camera API

The EIH Camera API includes two essential .proto files for setting up a gRPC client: the Camera.proto and the CameraAPI.proto. These files define the communication protocols and structure required for interacting with the EIH Camera, allowing developers to configure gRPC-based applications easily for controlling and accessing camera functions.

3.1 Camera.proto

Camera.proto defines the data structure used by the EIH Camera API.

3.1.1 CameraMatrix

Syntax

```
message CameraMatrix
{
    float matrix_00 = 1;
    float matrix_01 = 2;
    float matrix_02 = 3;
    float matrix_10 = 4;
    float matrix_11 = 5;
    float matrix_12 = 6;
    float matrix_20 = 7;
    float matrix_21 = 8;
    float matrix_22 = 9;
}
```

Description

Represents the camera matrix (3x3) of the EIH Camera, which defines how a point in 3D space will project onto the camera's 2D image plane.

3.1.2 DistortionCoefficients

Syntax

```
message DistortionCoefficients
{
    float coefficient_00 = 1;
    float coefficient_10 = 2;
    float coefficient_20 = 3;
    float coefficient_30 = 4;
    float coefficient_40 = 5;
}
```

Description

Represents the distortion coefficients (5x1) of the EIH Camera, used to correct distortions introduced by the camera lens during image capture.

3.1.3 HandEyeArray

Syntax

```
message HandEyeArray
{
    float handeye_x = 1;
    float handeye_y = 2;
    float handeye_z = 3;
    float handeye_rx = 4;
    float handeye_ry = 5;
    float handeye_rz = 6;
}
```

Description

Represents the Hand-Eye array (6x1), which determines the spatial relationship between the robot's end-effector and the EIH camera.

3.1.4 CaptureSetting_value

Syntax

```
message HandEyeArray
{
    int64 CurrentValue = 1;
    int64 MinValue = 2;
    int64 MaxValue = 3;
}
```

Description

Indicates the format of values used in various Capture Settings, which includes the current value and its range.

Parameters

CurrentValue	Current value of the specific capture value
Min Value	Lower boundary of the specific capture value
MaxValue	Upper boundary of the specific capture value

3.1.5 WhiteBalance

Syntax

```
message WhiteBalance
{
    CaptureSetting_value red_ratio = 1;
    CaptureSetting_value green_ratio = 2;
    CaptureSetting_value blue_ratio = 3;
}
```

Description

Indicates the white balance setting of the EIH Camera. White balance in a camera is the process of adjusting the colors in an image so that white appears true white, ensuring other colors are showing accurately. It compensates for the color temperature from the light source
and affects how to capture color.

Parameters

red_ratio	Adjustment of the red light intensity in an image.
green_ratio	Adjustment of the green light intensity in an image.
blue_ratio	Adjustment of the blue light intensity in an image.

3.1.6 Information

Syntax

```
message Information
{
    string SerialNumber = 1;
}
```

Description

A member of the class Camera. Reserved for future usage.

3.1.7 Intrinsics

Syntax

```
message Intrinsics
{
    float FocusValue = 1;
    int32 ImageWidth = 2;
    int32 ImageHeight = 3;
```

```
        CameraMatrix CameraMatrix = 4;
        DistortionCoefficients DistortionCoefficients = 5;
    }
```

Description

A member of the class Camera. Represents the internal parameters of the EIH Camera.

Parameters

FocusValue	Focus value used when the camera matrix is calibrated.
ImageWidth	Width of the image
ImageHeight	Height of the image
CameraMatrix	Camera matrix of the EIH Camera
DistortionCoefficients	Distortion coefficients of the EIH Camera

3.1.8 HandEyeParameters

Syntax

```
    message HandEyeParameters
    {
        HandEyeArray HandEyeArray = 1;
    }
```

Description

A member of the class Camera, which represents the spatial relationship between the robot's end effector and the EIH camera.

3.1.9 Image

Syntax

```
    message Image
    {
        message Data
        {
            bytes EncodeString = 1;
        }
        message Configuration
        {
            string ImageType = 1;
            string ImageSize = 2;
            int32 ImageWidth = 3;
            int32 ImageHeight = 4;
            string PixelFormat = 5;
        }
    }
```

Description

A member of the class Camera. Represents the image captured at the moment.

Parameters

Data	Member of the class Image, represents the image data captured.
EncodeString	The image captured, represents in encoded string format.
Configuration	Member of the class Image, represent of the configuration the image captured.
ImageType	The graphics file format of how the image is captured. The EIH camera currently only supports png format.
ImageSize	Size of the image captured, either 1M(1280*960) or 5M (2592*1944).
ImageWidth	Width of the image captured.
ImageHeight	Height of the image captured.
PixelFormat	Represents how the image's pixels are organized, either MONO or RGB

3.1.10 CapturingSettings

Syntax

```
message CapturingSettings
{
    CaptureSetting_value ShutterTime = 1;
    CaptureSetting_value Gain = 2;
    WhiteBalance WhiteBalance = 3;
    CaptureSetting_value Focus = 4;
    string ImageSize = 5;
}
```

Description

A member of the class Camera. Represents all sorts parameters of how the image is captured.

Parameters

ShutterTime	Refers to the amount of time (μs) the camera's sensor is exposed to light.
Gain	Refers to the brightness amplification (dB) setting.
WhiteBalance	Refers to the white balance setting.
Focus	Refers to the focus value of the camera.
ImageSize	Refers to the image size setting, which should be either 1M (1280*960) or 5M (2592*1944).

3.1.11 Camera

Syntax

```
message Camera
{
    message Information
```

```
    message Instrinsic
    message HandEyeParameters
    message Image
    message CaptureSettings

    repeated Intrinsics intrinsics = 1;
    HandEyeParameters handeyeParameters = 2;
    CapturingSettings capturingSettings = 3;
    Image image = 4;
}
```

Description

Represents the complete data of the EIH camera.

Parameters

intrinsics	Refers to the instrinsic of EIH Camera, see 3.1.7. A EIH Camera usually has multiple intrinsics data.
handeyeParameters	Refers to the hand-eye parameters between the robot and the EIH Camera, see 3.13 and 3.18.
capturingSettings	Refers to the capturing settings, see 3.1.10.
image	Refers to the image captured, see 3.1.9.

3.2 CameraAPI.proto

CameraAPI.proto defines EIH API functions.

3.2.1 isCameraConnected

Syntax

```
rpc isCameraConnected(google.protobuf.Empty)
returns(isCameraConnectedResponse)

message isCameraConnectedResponse
{
    bool isCameraConnected = 1;
    string connection_message = 2;
}
```

Description

Checks if the EIH Camera API connects to the camera.

Parameters

None

Return

isCameraConnected	True denotes the API connects to the camera; otherwise, it returns False.
connection_message	Details about the situation of the connection.

IMPORTANT:
If the API connects to the Camera, the message returned would be:

- Camera connected successfully.

On the other hand, if disconnected, the message returned would be one of the following：

- Another handler is currently occupied. Please try calling the API again later.
- TM_EIH_Camera is not connected.
- Only supports TM_EIH_Camera.
- TM_EIH_Camera is currently unavailable. Please try calling the API again later.

3.2.2 getIntrinsics

Syntax

```
rpc getIntrinsics(google.protobuf.Empty)
returns(getIntrinsicsResponse)

message getIntrinsicsResponse
```

```
{
    repeated Camera.Intrinsics cam_intrinsics = 1;
}
```

Description

Gets the intrinsics of the EIH Camera. Usually, there should be around 13 intrinsics data, depending on the camera calibration method.

Parameters

None

Return

cam_intrinsics	Refers to the intrinsics of the camera. For details, refer to 3.1.7.

3.2.3 getHandEyeParameters

Syntax

```
rpc getHandEyeParameters (google.protobuf.Empty)
returns(Camera.HandEyeParameters)
```

Description

Gets the Eye-Hand parameter between the robot and the EIH Camera.

Parameters

None

Return

Camera.HandEyeParameters	For details of the Eye-Hand parameters between the robot and the EIH Camera, refer to 3.1.3 and 3.1.8.

3.2.4 getImageData

Syntax

```
rpc getImageData (google.protobuf.Empty)
returns(Camera.Image.Data)
```

Description

Requests the EIH Camera to capture and return an image.

Parameters

None

Return

Camera.Image.Data	Refers to the image captured by the EIH Camera, which is formatted as an encoded string. For details, refer to 3.1.9.

3.2.5 getImageConfiguration

Syntax

```
rpc getImageConfiguration (google.protobuf.Empty)
returns(Camera.Image.Configuration)
```

Description

Gets the configurations of image to be captured.

Parameters

None

Return

Camera.Image.Configuration	Configurations of image capturing of the EIH Camera. For details, refer to 3.1.9.

3.2.6 terminateCameraConnection

Syntax

```
rpc terminateCameraConnection (google.protobuf.Empty)
returns(google.protobuf.Empty)
```

Description

Disconnects the EIH Camera from the robot. In such cases, neither a vision job editing or running TMvision nor the EIH Camera API can exploit the camera.

Parameters

None

Return

None

IMPORTANT

IMPORTANT:
Under normal circumstances, users are advised not to use the terminateCameraConnection function, as it is intended for a more advanced vision system integration with the EIH Camera and is not an official feature yet. For additional information, please reach Techman Robot Inc.

3.2.7 resumeCameraConnection

Syntax

```
rpc resumeCameraConnection (google.protobuf.Empty)
returns(google.protobuf.Empty)
```

Description

Resumes the conntection between the EIH Camera and the robot.

Parameters

None

Return

None

3.2.8 getCapturingSettings

Syntax

```
rpc getCapturingSettings (google.protobuf.Empty)
returns(Camera.CapturingSettings)
```

Description

Gets the capturing settings.

Parameters

None

Return

Camera.CapturingSettings	Settings of how the image to be captured. For details, refer to 3.1.10.

3.2.9 setCapturingSettings

Syntax

```
rpc setCapturingSettings (setCapturingSettingsRequest)
returns(google.protobuf.Empty)

message setCapturingSettingsRequest
{
    int64 ShutterTime = 1;
    int64 Gain = 2;
    int64 wb_redratio = 3;
    int64 wb_greenratio = 4;
    int64 wb_blueratio = 5;
```

```
    int64 Focus = 6;
    int64 ImageSize = 7;
}
```

Description

Sets the capturing settings.

Parameters

ShutterTime	Denotes the amount of time the camera's sensor is exposed to light.
Gain	Denotes the gain (brightness amplification) setting.
wb_redratio	Denotes the red ratio of the white balance setting, i.e., the adjustment of the red light intensity in an image.
wb_greenratio	Denotes the green ratio of the white balance setting, i.e., the adjustment of the green light intensity in an image.
wb_blueratio	Denotes the blue ratio of the white balance setting, i.e., the adjustment of the blue light intensity in an image.
Focus	Denotes the focus value setting of the camera when capturing an image.
ImageSize	Denotes the image size setting, which should be either 1M (1280*960) or 5M (2592*1944).

Return

None

3.2.10 getShutterTime

Syntax

```
rpc getShutterTime (google.protobuf.Empty)
returns(getShutterTimeResponse)

message getShutterTimeResponse
{
    CaptureSetting_value shuttertime = 1;
}
```

Description

Gets the shuttle time setting.

Parameters

None

Return

shuttertime	The current shuttle time value along with the boundadry values.

3.2.11 setShutterTime

Syntax

```
rpc setShutterTime (setShutterTimeRequest)
returns(google.protobuf.Empty)

message setShutterTimeRequest
{
    int64 shuttertime = 1;
}
```

Description

Sets the shuttle time.

Parameters

shuttertime	Denotes the amount of time the camera's sensor is exposed to light.

Return

None

3.2.12 getGain

Syntax

```
rpc getGain (google.protobuf.Empty)
returns(getGainResponse)

message getGainResponse
{
    CaptureSetting_value gain = 1;
}
```

Description

Gets the gain setting.

Parameters

None

Return

shuttertime	The current gain value along with the boundadry values.

3.2.13 setGain

Syntax

```
rpc setGain (setGainRequest)
returns(google.protobuf.Empty)

message setGainRequest
{
    int64 gain = 1;
}
```

Description

Sets the gain.

Parameters

gain	Refers to the gain (brightness amplification) setting.

Return

None

3.2.14 getWhiteBalance

Syntax

```
rpc getWhiteBalance (google.protobuf.Empty)
returns(getGainResponse)

message getWhiteBalanceResponse
{
    WhiteBalance whitebalance = 1;
}
```

Description

Gets the white balance setting.

Parameters

None

Return

whitebalance	The current gain value along with the boundadry values. For details, refer to 3.1.5.

3.2.15 setWhiteBalance

Syntax

```
rpc setWhiteBalance (setWhiteBalanceRequest)
returns(google.protobuf.Empty)

message setWhiteBalanceRequest
{
    int64 wb_redratio = 1;
    int64 wb_ greenratio = 2;
    int64 wb_ blueratio = 3;
}
```

Description

Sets the white balance setting.

Parameters

wb_redratio	Denotes the red ratio of the white balance setting, i.e., the adjustment of the red light intensity in an image.
wb_ greenratio	Denotes the green ratio of the white balance setting, i.e., the adjustment of the green light intensity in an image.
wb_ blueratio	Denotes the blue ratio of the white balance setting, i.e., the adjustment of the blue light intensity in an image.

Return

None

3.2.16 getFocus

Syntax

```
rpc getFocus (google.protobuf.Empty)
returns(getFocusResponse)

message getFocusResponse
{
    CaptureSetting_value focus = 1;
}
```

Description

Gets the focus setting.

Parameters

None

Return

focus The current focus value of the camera along with the boundadry values.

3.2.17 setFocus

Syntax

```
rpc setFocus (setFocusRequest)
returns(google.protobuf.Empty)

message setGainRequest
{
    int64 focus = 1;
}
```

Description

Sets the focus.

Parameters

focus Denotes the focus value of the camera.

Return

None

3.2.18 getImageSize

Syntax

```
rpc getImageSize (google.protobuf.Empty)
returns(getImageSizeResponse)

message getImageSizeResponse
{
    string imagesize = 1;
}
```

Description

Gets the image size setting.

Parameters

None

Return

imagesize	The current image size setting.

3.2.19 setImageSize

Syntax

```
rpc setImageSize (setImageSizeRequest)
returns(google.protobuf.Empty)

message setImageSizeRequest
{
    string imagesize = 1;
}
```

Description

Sets the image size.

Parameters

focus	Denotes the image size setting, which should be either 1M (1280*960) or 5M (2592*1944).

Return

None

4. Error Handling

After requesting the EIH Camera API, users should verify the grpc status and retrieve the error message if needed. Below is an example of how to use the C++ grpc library function for this purpose:

```
grpc::Status status = m_Stub->isCameraConnected(&context, request, &response);
if (status.ok())
{
   std::cout << "Camera connection status: " << response.iscameraconnected() << std::endl;
   std::cout << "Connection message: " << response.connection_message() << std::endl;
}
else
{
   std::cout << "RPC failed: " << status.error_code() << ": " << status.error_message() << std::endl;
}
```

The possible error message would be one of the following:

- *Another handler occupies currently. Please try calling the API again later.*
  This message indicates that another source is currently using the EIH camera. For example, TMvision is running a vision job, or another client is occupying the EIH Camera API.

- *TM_EIH_Camera is not connected.*
  This message means the EIH camera is not connected currently. Please resume the connection before using the EIH Camera API.

- *Only supports TM_EIH_Camera.*
  This message means there is no EIH Camera on the camera list.

- *TM_EIH_Camera is currently unavailable. Please try calling the API again later.*
  This message means the EIH camera is unavailable. For example, the camera has not been calibrated yet.

- *Setting failed due to invalid parameter.*
  This message means the recent request does not provide the correct parameters; for example, out of range.

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